No of Pages : 3 **Course Code : 23MX14**

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, JUNE 2024

MCA Semester : 1

23MX14 DATABASE SYSTEMS

Time : 3 Hours **Maximum Marks : 100**

INSTRUCTIONS:
1. Answer **ALL** questions. Each question carries 25 Marks.
2. **In each question, Section a carries 3 Marks, Section b carries 10 Marks and Section c carries 12 Marks**

1. a) What is data redundancy? Can data redundancy be completely eliminated? Justify your answer.

   b) i) With a neat diagram, explain how data is accessed from a database?(5)

   ii) Consider you want to build a video site similar to YouTube. The actual video data, the metadata about the video, such as title, the user who uploaded it must be stored and video can be accessed by users simultaneously. Is it better to use a file processing system or a database for storage? What problems do you foresee? Justify. [5]

   c) A real estate firm lists property for sale. The following points describe the organization:

   - The firm has a number of sales offices in several states. Attributes of sales office include office_number (identifier) and address, city, state and zip.
   - Each sales office is assigned with one or more employees. Attributes of employee include employee_id (identifier) and employee_name. An employee must be assigned to only one sales office.
   - For each sales office, there is always one employee assigned to manage that office. An employee may manage only the sales office to which he or she is assigned.
   - The firm lists property for sale. Attributes of property include property_id (identifier), and address, city, state and zip.
   - Each unit of property must be listed with one and only one of the sales offices. A sales office may have any number of properties listed or may have no properties listed.
   - Each unit of property has one or more owners. Attributes of owner are owner_id (identifier) and owner name. An owner may own one or more units of property.
   - The firm wants to keep track of the percentage that a given owner owns a given piece of property. For example, imagine that Smith and Jones both own property A. Smith owns 35% of property A and Jones owns 65% of property A. The firm wants to retain this information and they want to call that attribute percent_owned.

   Using the above details construct the ER Diagram by identifying

i. the Entity and its relevant attributes.

ii. the primary key, Weak Entity and Strong Entity.

iii. the relationship between the entity types.

iv. the Connectivity and Cardinality on relationship.

2. a) What is data independence? What are the differences between physical and logical data independence?

b) i) What do you understand by union compatibility? Why do UNION, INTERSECTION, and DIFFERENCE operations require the relations on which they are applied to be union compatible? [5]

ii) Explain the significance of enforcing integrity constraints in a relational data model with suitable examples. [5]

c) i) Explain the various types of JOIN operations?

Consider the following relational schema

Student (rno,name,dept,age)

Books(isbn,title,authors,publisher)

borrow(rno, isbn,date)

Write the following queries in relational algebra.

a. Find the names of student who have borrowed a book Published by OReily?

b. Find the names of student who have borrowed all books Published by OReily?

c. Find the names of student who have borrowed more than five different books published by OReily?

d. For each publisher, find the names of students who have borrowed?

(OR)

ii) Explain the operations UNION, INTERSECTION, SELECTION and PROJECTION in relational algebra. Consider the following relational schemas. Express each of the provided information needs as a relational algebra query using only projection (π), selection (σ), renaming (ρ), inner joins , set union/intersection (∪, ∩), and division (/).

Patient(aadhar_no:integer, name:string, address:string, age:integer)

Doctor(aadhar_no:integer, name:string, specialty:string, experience:integer)

PrimaryPhysician(patient_aadhar:integer,doctor_aadhar:integer, firstAppointment:date)

i) Find the aadhar_no of doctors who see patients that are at least 25 years old.

ii) Find the aadhar_no of patients that see a doctor that has more than 20 years of experience or less than 2 years of experience.

3. a) Differentiate between single row functions and aggregate functions with simple examples?

   b) Explain any 5 data manipulation commands of SQL with suitable examples.

   c) Define the employee table structure using SQL create command with the following constraints. Name is a primary key; Gender has only two possible values; Skill1 and Skill2 have null values; Designation, Salary, DoB and DoJ are NotNull. Salary range is from 10000 to 45000.

Name	Design	Salary	Gender	DoB	DoJ	Skill1	Skill2
Arul	Developer	17000	M	21-Apr-66	10-Jun-86	Pascal	Cobol
Anand	Cobol Programmer	15000	M	11-Sep-75	17-Jun-97	Cobol	Basic
Sumathi	System Analyst	21000	F	07-Feb-66	21-Apr-91	Cobol	Basic

   Write SQL queries and its equivalent Relational Algebra expression for the following requirements:

   i) List out the Female programmers getting salary in the range of 14000 to 25000.

   ii) Display the highest, lowest and average salaries for those earning more than 17000.

   iii) Find the number of employees getting salary greater than or equal to 20000.

   iv) Display employees who were joined to the company before year 1991.

   v) Display the details of all employees as below as below using SQL.

   "Dear Arul, you are working as Developer by getting Rs.17000".

4. a) Give a set of functional dependencies (FD)s for the relational schema R(A, B, C, D) with primary key AB under which R is in 1NF but not in 2NF.

   b) Explain how various stages of normalization are related to functional dependencies.

   c) Consider the following relations for published books:

   Book (Book_title, Author_name, Book_type, List_price, Author_ affiliation, Publisher)

   Suppose the following dependencies exist:

   Book_title $\rightarrow$ Publisher, Book_type

   Book_type $\rightarrow$ List_price

   Author_name $\rightarrow$ Author_affiliation

   1) In what normal form is the relation? Explain your answer.

   2) Apply normalization until you cannot decompose the relations further. State the reasons behind each decomposition.

/END/

FD/RSP